

System Design and Software Architecture

AI Study and Task Planner

This document presents the technical design, architecture, database design, user interaction flow, activity flow, API design, and AI integration workflow for the proposed system.

Information	Details
Course Code	CSE4204
Course Title	Mobile Computing Lab
Section	8B
Team Number	T03
Official Team Name	CSE4204-8B-T03
Project Title	AI Study and Task Planner
Repository Link	https://github.com/saimhasan10/CSE4204-8B-t03-ai-study-and-task-planner

Team Information

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1. Project Information

AI Study and Task Planner is a web-based system for university students to manage study tasks, project work, deadlines, and academic planning in one place. The system will allow students to create tasks, set deadlines, select priority levels, track progress, and receive AI-based planning support.

The main purpose of this system design document is to convert the project requirements into a clear technical blueprint. It explains how different modules will communicate, how data will be stored, how users will interact with the system, and how AI will be integrated through the backend.

1.1 Objectives of the Design

- To define the overall structure of the system before implementation.
- To explain how frontend, backend, database, and AI service will communicate.
- To provide a database design that supports task, course, AI plan, and chat data.
- To define user interactions through updated use cases and an activity workflow.
- To prepare planned API endpoints for development using Django REST Framework.
- To explain how Gemini API will provide AI-based study planning support.

1.2 Technology Stack

Part	Technology
Frontend	React.js
Styling	Tailwind CSS
Backend	Django
API Framework	Django REST Framework
Database	PostgreSQL
AI Service	Gemini API
Frontend Deployment	Vercel / Netlify
Backend Deployment	Render / Railway
Database Hosting	Supabase / Neon
Version Control	GitHub

2. System Architecture Diagram

The architecture diagram shows the main technical components of the system. The student uses the React frontend through a browser. The frontend sends HTTPS API requests to the Django REST API backend. The backend stores and retrieves data from PostgreSQL and connects with Gemini API for AI-based study planning features.

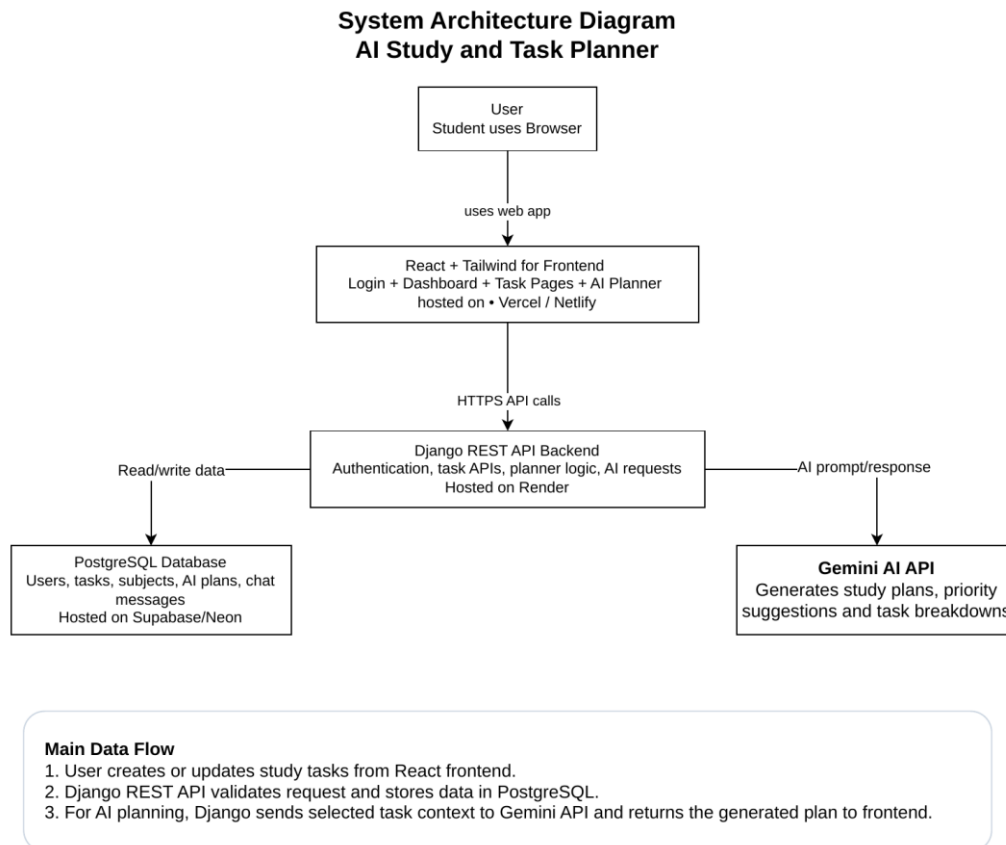


Figure 1: System Architecture Diagram

3. ER Diagram and Database Design

The ER diagram defines the main database entities, attributes, primary keys, foreign keys, and relationships. The updated Task entity includes course_id as an optional foreign key so that study tasks can be connected to a specific course or subject.

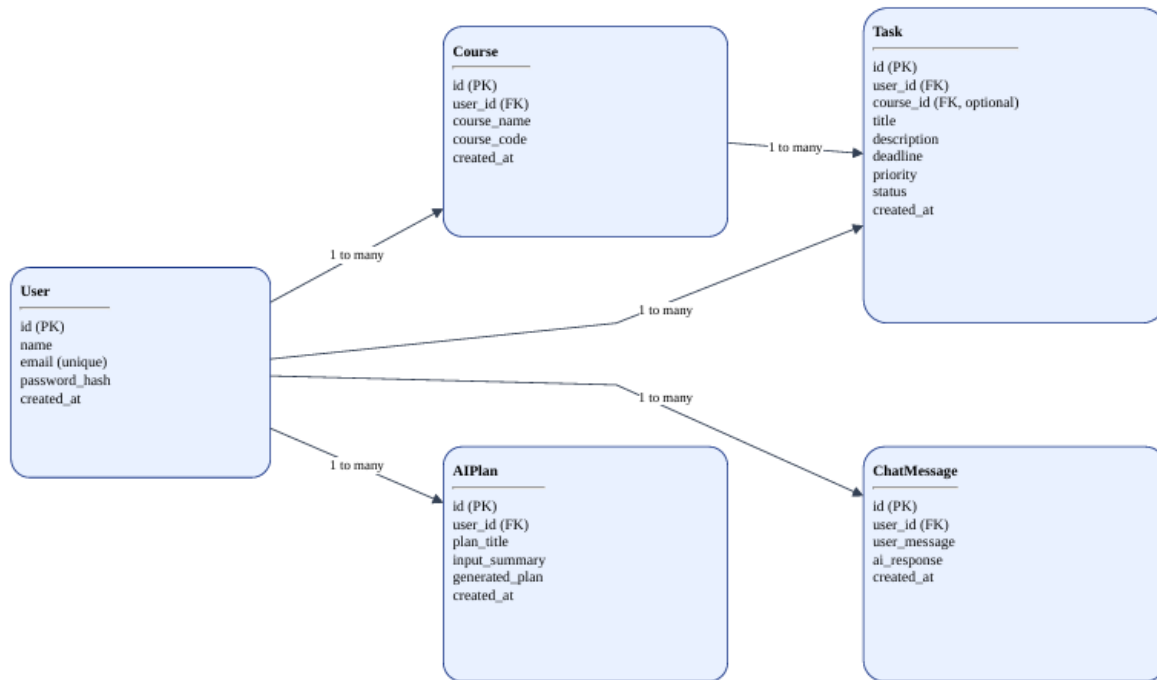


Figure 2: ER Diagram

3.1 Main Database Entities

Entity	Purpose
User	Stores student account information such as name, email, password hash, and account creation time.
Course	Stores course or subject information created by the user.
Task	Stores study tasks with deadline, priority, status, user reference, and optional course reference.
AIPlan	Stores AI-generated study plans and the related input summary.
ChatMessage	Stores user messages and AI responses for study/task-related chat.

4. Updated Use Case Diagram

The updated use case diagram shows the major functionalities of the system. The main actor is the Student/User, who can manage tasks, track progress, and use AI-based features. The Admin can view registered users and monitor basic system activity.

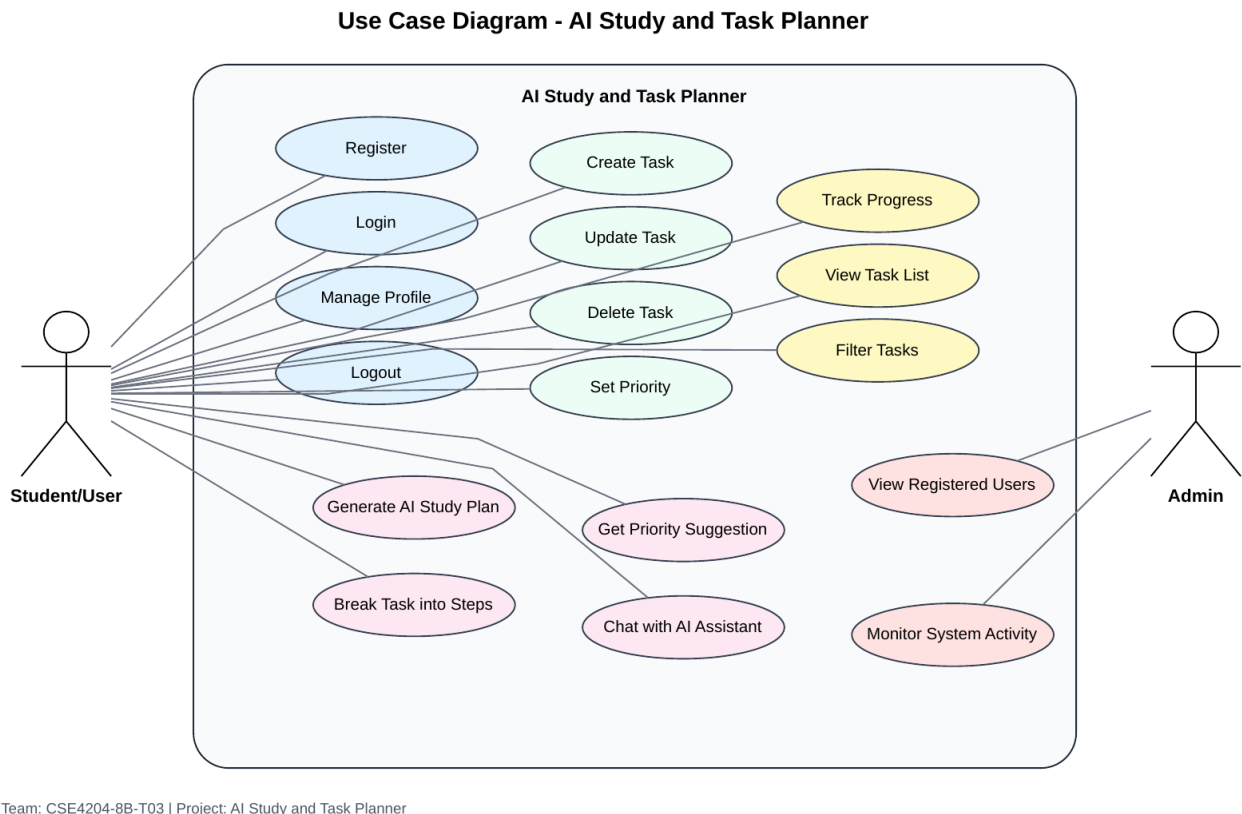


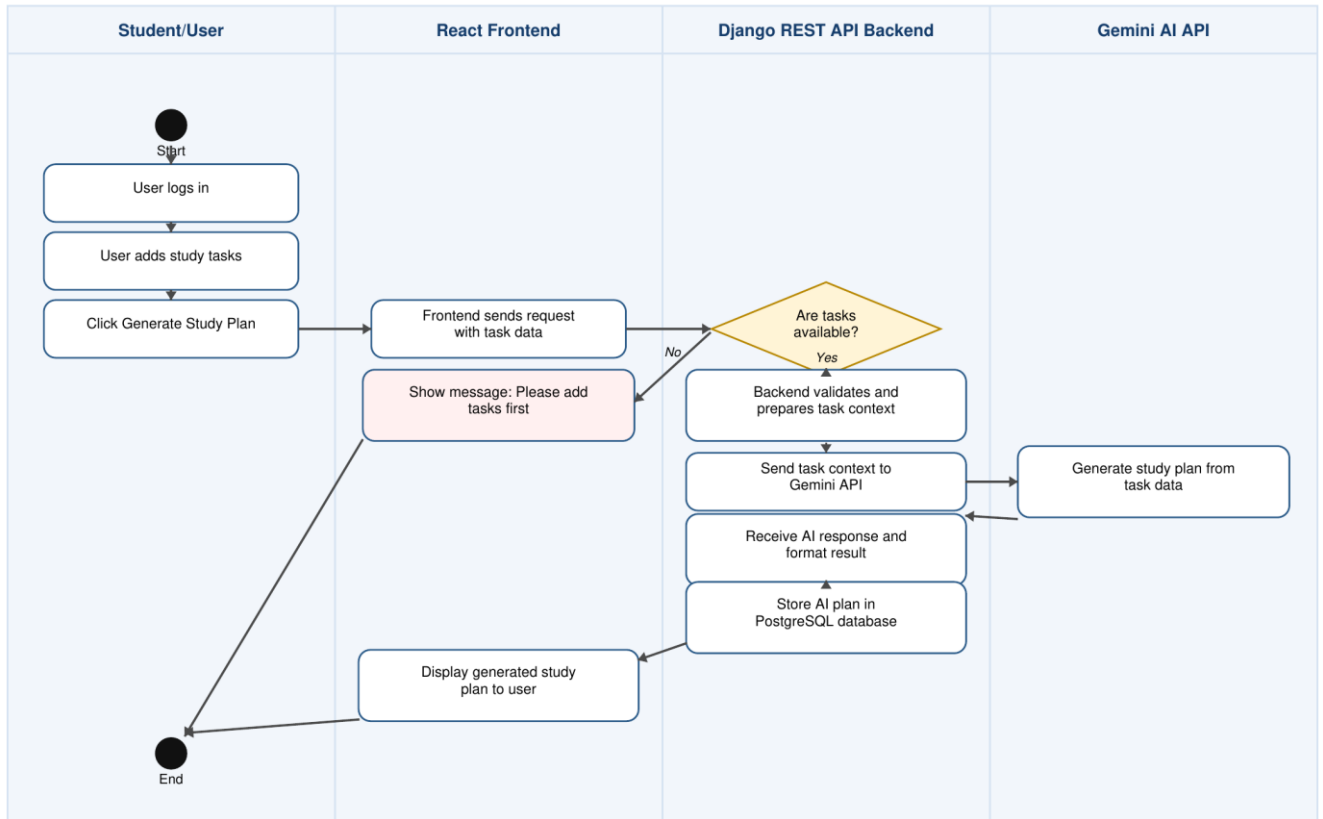
Figure 3: Use Case Diagram

5. Activity Diagram

The activity diagram shows the workflow of the AI Study Plan Generation feature. It includes user actions, frontend request, backend validation, decision point, Gemini API processing, database storage, and final output display.

Activity Diagram - AI Study Plan Generation Workflow

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Figure 4: Activity Diagram - AI Study Plan Generation Workflow

6. API Design

The backend API will be developed using Django REST Framework. The frontend will communicate with these endpoints for authentication, profile management, course management, task management, progress tracking, and AI-based planning features.

Method	Endpoint	Purpose	Input Data	Expected Output
POST	/api/register/	Register a new user	Name, email, password	Account created message and user information
POST	/api/login/	Login registered user	Email, password	Login success message and authentication token/session
POST	/api/logout/	Logout current user	User token/session	Logout success message
GET	/api/profile/	View user profile	User token/session	User profile details
PUT	/api/profile/	Update user profile	Name, email, updated profile data	Updated profile information
GET	/api/courses/	View all courses/subjects	User token/session	List of courses
POST	/api/courses/	Add a new course/subject	Course name, course code	New course created message
PUT	/api/courses/{id}/	Update course information	Course name, course code	Updated course information
DELETE	/api/courses/{id}/	Delete a course	Course ID	Course deleted message
GET	/api/tasks/	View all study tasks	User token/session	List of user tasks
POST	/api/tasks/	Create a new study task	Title, description, deadline, priority, status, course ID	New task created message
GET	/api/tasks/{id}/	View a single task	Task ID	Task details
PUT	/api/tasks/{id}/	Update task information	Updated title, description, deadline, priority, status	Updated task information
DELETE	/api/tasks/{id}/	Delete a task	Task ID	Task deleted message
GET	/api/tasks/filter/	Filter tasks	Priority/status/deadline query	Filtered task list
GET	/api/progress/	View progress overview	User token/session	Total tasks, completed tasks, pending tasks, progress percentage
POST	/api/ai/generate-study-plan/	Generate AI study plan	Task list, deadline, priority, status	AI-generated study plan
POST	/api/ai/suggest-priority/	Suggest task priority	Task title, deadline, priority, description	AI priority suggestion
POST	/api/ai/break-task/	Break a large task into smaller steps	Task title and description	Step-by-step task breakdown
POST	/api/ai/chat/	Study/task-related AI chat	User message	AI response
GET	/api/ai/plans/	View saved AI plans	User token/session	List of saved AI plans
GET	/api/ai/plans/{id}/	View a specific AI-generated plan	Plan ID	AI plan details

7. AI Integration Workflow

7.1 Why AI is Needed

AI is needed because students may have many study tasks, deadlines, assignments, lab reports, and project work at the same time. A normal task management system can store tasks, but it cannot guide students on how to plan work effectively. The AI assistant will generate study plans, suggest task priority, and break large tasks into smaller steps.

7.2 AI Service / Model

The system will use Gemini API as the AI service. Gemini API will be connected through the Django REST API backend. The React frontend will not directly call the Gemini API, which helps keep the API key secure.

7.3 Input and Output of AI System

Type	Details
Input to AI	Task title, task description, deadline, priority, status, and related course/subject information.
Output from AI	AI-generated study plan, task priority suggestion, task breakdown, and short study guidance.

7.4 AI Workflow Steps

- Student/User logs in to the system.
- Student creates study tasks with deadline, priority, and status.
- Student clicks Generate Study Plan from the frontend.
- React frontend sends the request to the Django REST API backend.
- Backend collects selected task data from PostgreSQL.
- Backend sends prepared task context to Gemini API.
- Gemini API generates a study plan or suggestion.
- Backend stores the generated response in the AIPlan or ChatMessage table.
- React frontend displays the generated result to the student.

7.5 How AI Improves the Project

AI improves the project by adding planning support instead of only storing tasks. It helps students understand which task is more urgent, how to divide large tasks, and how to follow a practical study plan. This makes the system more useful for academic productivity and reduces last-minute pressure.

8. GitHub Repository Progress

The GitHub repository is maintained for documentation, diagrams, and future development. The repository link is:

<https://github.com/saimhasan10/CSE4204-8B-t03-ai-study-and-task-planner/tree/main>

The planned repository structure is shown below:

```
CSE4204-8B-t03-ai-study-and-task-planner/  
|  
├─ backend/  
├─ frontend/  
├─ database/  
├─ documentation/  
|   ├─ proposal/  
|   ├─ srs/  
|   └─ system-design/  
├─ diagrams/  
|   ├─ use-case/  
|   ├─ er-diagram/  
|   ├─ architecture/  
|   └─ activity-diagram/  
├─ README.md  
└─ .gitignore
```

9. References

1. Django Documentation - <https://docs.djangoproject.com/>
2. Django REST Framework Documentation - <https://www.django-rest-framework.org/>
3. React Documentation - <https://react.dev/>
4. Tailwind CSS Documentation - <https://tailwindcss.com/docs>
5. PostgreSQL Documentation - <https://www.postgresql.org/docs/>
6. Gemini API Documentation - <https://ai.google.dev/>
7. Draw.io / Diagrams.net - <https://app.diagrams.net/>